**Suggested problems:** 1, 3, 6, 8

---

**Problem 1:** (Example 3.7.5) Suppose that $n$ customers arrive at a single-server queue for service. Let $X_i$ be the amount of time that the server spends serving customer $i$ for $i = 1, 2, \ldots, n$. The $n$-dimensional random vector $\mathbf{X} = (X_1, X_2, \ldots, X_n)$ has density function

$$f(\mathbf{x}) = \begin{cases} \dfrac{c}{(2 + \sum_{i=1}^{n} x_i)^{n+1}} & : \text{for all } x_i > 0, \\ 0 & : \text{else,} \end{cases}$$

for some positive constant $c$.

(a) Compute the value of $c$.

(b) Suppose $n = 5$, and compute the marginal density function $f(x_1, x_4)$.

(c) Continuing with $n = 5$, compute the marginal density function $f(x_2)$.

(d) Continuing again with $n = 5$, compute $P(X_2 > 3)$.

**Problem 2:** Assuming $n = 5$, are the random variables $X_1$, $X_2$, and $X_3$ in Problem 1 independent?